

COURSE INFORMATION SHEET

Course code: PH24101

Course title: PHYSICS

Pre-requisite(s): Intermediate Physics and Intermediate Mathematics

Co- requisite(s): Mathematics I

Credits: L:3 T:1 P:0

Class schedule per week: 4

Class: B. Tech

Semester / Level: I

Branch: ALL

Name of Teacher:

Course Objectives

This course enables the students:

A.	To explain principles of physical optics and to have basic idea of fiber optics.
B.	To construct Maxwell's equations from basic principles and use it to solve electromagnetic plane wave equations.
C.	To distinguish between Newtonian Mechanics and special theory of relativity and develop the relationship of length contraction, time dilation and Einstein energy mass relation and to apply the concepts of special theory of relativity in various field of physics and engineering.
D.	To illustrate the phenomena of old quantum theory and derive Heisenberg uncertainty principle and Schrodingers equations
E.	To interpret basic lasing action, study various types of lasers, and to have basic idea of nuclear physics and plasma physics

Course Outcomes

After the completion of this course, students will be:

1.	Able to analyse the intensity variation of light due to Polarization, interference and diffraction.
2.	Able to formulate and solve the engineering problems on electromagnetism
3.	Able to explain special theory of relativity and apply its concepts in various fields of physics and engineering.
4.	Able to explain fundamentals of quantum mechanics and apply it to problems on bound states
5.	Able to explain working principle of lasers and to summarize its applications, understand basic idea of nuclear and plasma physics

Physics for 1st Semester B.Tech (CBCS)

Course Code: PH24101

Title : PHYSICS

Module-1: Physical Optics: Polarization, Malus' Law, Brewster's Law, Double Refraction, Interference in thin films (Parallel films), Interference in wedge-shaped layers, Newton's rings, Fraunhofer diffraction by single slit, Double slit. Elementary ideas of fibre optics and application of fibre optic cables. [8]

Module-2: Electromagnetic Theory: Gradient, Divergence and Curl, Statement of Gauss theorem & Stokes theorem, Gauss's law, Applications, Concept of electric potential, Relationship between E and V, Polarization of dielectrics, dielectric constant, Boundary conditions for E & D, Gauss's law in magnetostatics, Ampere's circuital law, Boundary conditions for B & H, Equation of continuity, Displacement current, Maxwell's equations. [8]

Module-3: Special Theory of Relativity: Introduction, Inertial frame of reference, Galilean transformations, Postulates, Lorentz transformations and its conclusions, Length contraction, time dilation, velocity addition, Mass change, Einstein's mass energy relation. [6]

Module-4: Quantum Mechanics: Planck's theory of black-body radiation, Compton effect, Wave particle duality, De Broglie waves, Davisson and Germer's experiment, Uncertainty principle, Brief idea of Wave Packet, Wave Function and its physical interpretation, Schrodinger equation in one-dimension, free particle, particle in an infinite square well. [9]

Module-5: Modern Physics: Laser-Spontaneous and stimulated emission, Einstein's A and B coefficients, Population inversion, Light amplification, Basic laser action, Ruby and He-Ne lasers, Properties and applications of laser radiation, Nuclear Physics- Binding Energy Curve, Nuclear Force, Liquid drop model, Introduction to Shell model, Applications of Nuclear Physics, Concept of Plasma Physics, and its applications. [9]

Text books:

- 1: A. Ghatak, Optics, 4th Edition, Tata Mcgraw Hill, 2009
- 2: Mathew N.O. Sadiku, Elements of Electromagnetics, Oxford University Press (2001)
- 3: Arthur Beiser, Concept of Modern Physics, 6th edition 2009, Tata McGraw- Hill
4. F. F. Chen, Introduction to Plasma Physics and controlled Fusion, Springer, Edition 2016.

Reference books:

- 1: Fundamentals of Physics, Halliday, Walker and Resnick

Gaps in the syllabus (to meet Industry/Profession requirements) : NA

POs met through Gaps in the Syllabus : NA

Topics beyond syllabus/Advanced topics/Design : NA

POs met through Topics beyond syllabus/Advanced topics/Design

Course Delivery methods	
Lecture by use of boards/LCD projectors/OHP projectors	Y
Tutorials/Assignments	Y
Seminars	N
Mini projects/Projects	N
Laboratory experiments/teaching aids	N
Industrial/guest lectures	N
Industrial visits/in-plant training	N
Self- learning such as use of NPTEL materials and internets	Y
Simulation	N

Course Outcome (CO) Attainment Assessment tools & Evaluation procedure

Direct Assessment

Assessment Tool	% Contribution during CO Assessment
Mid Sem Examination Marks	25
End Sem Examination Marks	50
Quiz	10+10
Teacher's assessment	5

Assessment Compoents	CO1	CO2	CO3	CO4	CO5
Mid Sem Examination Marks	√	√	√		
End Sem Examination Marks	√	√	√	√	√
Quiz I			√	√	
Quiz II				√	√

Indirect Assessment –

1. Student Feedback on Faculty
2. Student Feedback on Course Outcome

Mapping between Objectives and Outcomes**Mapping between Course Objectives and Course Outcomes**

Course Objectives	1	2	3	4	5
A	H	M	M	L	H
B	M	H	M	L	H
C	L	L	H	L	L
D	-	L	L	H	H
E	H	M	L	L	H

Mapping of Course Outcomes onto Program Outcomes

Course Outcome #	Program Outcomes											
	a	b	c	d	e	f	g	h	i	j	k	l
1	H	M		H	H	H		H	M	M		H
2	H	H		H	H	H		H	M	M		H
3	H	L		M	L	M		H	M	M		H
4	H			H	M	M		M	M	M		H
"HoD Physics, BIT Mesra, Ranchi"<hod.phy@bitmesra.ac.in> 5	M	H		H	H	H	H	M	M	M		H

Mapping Between COs and Course Delivery (CD) methods

CD	Course Delivery methods	Course Outcome	Course Delivery Method
CD1	Lecture by use of boards/LCD projectors/OHP projectors	CO1	CD1 and CD2
CD2	Tutorials/Assignments	CO2	CD1 and CD2
CD3	Seminars	CO3	CD1 and CD2
CD4	Mini projects/Projects	CO4	CD1 and CD2
CD5	Laboratory experiments/teaching aids	CO5	CD1 and CD2
CD6	Industrial/guest lectures	-	-
CD7	Industrial visits/in-plant training	-	-
CD8	Self- learning such as use of NPTEL materials and internets	-	-
CD9	Simulation	-	-

Lecture wise Lesson planning Details.

Week No.	Lect. No.	Tentative Date	Ch. No.	Topics to be covered	Text Book / References	COs mapped	Actual Content covered	Methodology used	Remarks by faculty if any
1	L1-L2		1	Polarization, Malus' Law, Brewster's Law	T1, R1				
	L3-L5			Double Refraction, Interference in thin films (Parallel films), Interference in wedge-shaped layers	T1				
	L6			Newton's rings	T1				
	L7			Fraunhofer diffraction by single slit	T1, R1				
	L8			Double slit, Fiber optics introduction and its application	T1				
	L9			Gradient, Divergence, Curl and their physical interpretation, Gauss theorem, Stokes theorem, Gauss's law	T2, R1				
	L10-11			Concept of electric potential, Relationship between E and V of Polarization of Dielectrics, dielectric constant	T2, R1				
	L12-14			Boundary conditions for E & D , Equation of continuity Gauss's law in magnetostatics, Ampere's circuital law	T2				
	L15			Boundary conditions for B & H , Displacement current, equations, Maxwell's equations	T2				
	L16			Introduction, Inertial Frame of reference, Galilean transformations, Postulates	T2				
	L17-18				T3				

	L19-20			Lorentz transformations and its conclusions, Length contraction, time dilation,	T3				
	L21-22			Velocity addition, Mass change, Einstein's mass energy relation.	T3				
	L23-24			Planck's theory of black-body radiation, Compton effect	T3				
	L25-26			Wave particle duality, De Broglie waves Davisson and Germer's experiment	T3				
	L27-28			Uncertainty principle, Brief idea of Wave Packet, Wave Function and its physical interpretation	T3				
	L29-30			Schrodinger equation in one dimension	T3				
	L31-32			free particle particle in an infinite square well,	T3				
	L33-L34			Laser-Spontaneous and stimulated emission, Einstein's A and B coefficients, Population inversion, Light amplification	T1				
	L35-36			Basic laser action, Ruby and He-Ne lasers, Properties and applications of laser radiation	T1				
	L37-38			Binding Energy Curve, Nuclear Force, Liquid drop model, Introduction to Shell model	T3				
	L39-40			Applications of Nuclear Physics,	T3				
				Basic Concept of Plasma Physics and its applications					